

Comparative Analysis of Locally Produced Bitumen and Imported Low-Cost Bitumen for Bangladesh Road Construction (Focusing on Aggregate Shape)

by

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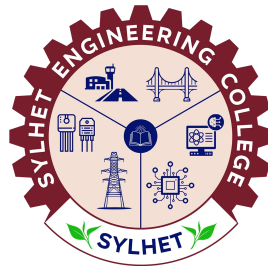
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Candidate's Declaration

This is to certify that the work presented in this thesis entitled, **“Comparative Analysis of Locally Produced Bitumen and Imported Low-Cost Bitumen for Bangladesh Road Construction (Focusing on Aggregate Shape)”**, is the outcome of the research carried out by the authors under the supervision of **Md Johir Bin Alam, Professor, Department of CEE, Shahjalal University of Science & Technology, Bangladesh**, and the co-supervision of **Mohiuddin Arif, Lecturer, Department of Civil Engineering, Sylhet Engineering College, Bangladesh**.

It is also declared that neither this thesis nor any part thereof has been submitted anywhere else for the award of any degree, diploma, or other qualifications.

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is a record of research work carried out under our supervision and we, hereby, approve that the report is submitted in partial fulfilment of the requirement for the award of their Bachelor’s Degree.

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Certificate of Acceptance

The thesis titled “**Comparative Analysis of Locally Produced Bitumen and Imported Low-Cost Bitumen for Bangladesh Road Construction (Focusing on Aggregate Shape)**” submitted by Md. Rakibul Hasan Mridha and Md Sagor Hossain Saddas; Student ID: 2020333545 and 2020333553, to the Department of Civil Engineering, Sylhet Engineering College, has been accepted as satisfactory in partial fulfillment of the requirement for the Degree of Bachelor of Science in Civil Engineering and approved as to its style and contents.

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The Authors

Abstract

Bangladesh's expanding road network relies predominantly on flexible pavements, in which bitumen serves as the principal binding agent. However, quality variations between domestically refined and imported low-cost bitumen, compounded by poorly characterised aggregate shape effects, contribute to premature pavement distress including rutting, stripping, and fatigue cracking. Despite the critical role these factors play, no prior study has systematically compared locally produced bitumen from Eastern Refinery with imported low-cost alternatives while simultaneously evaluating the influence of aggregate particle shape under conditions representative of Bangladesh's subtropical monsoon climate. This study addresses the gap through a laboratory-based comparative analysis employing the Marshall mix design method. Four distinct mixes were investigated: local bitumen with natural aggregates (control), imported bitumen with natural aggregates, local bitumen with 20% cubical aggregate replacement, and imported bitumen with 20% cubical aggregate replacement. Specimens were compacted at 75 blows per face in accordance with Roads and Highways Department heavy-traffic specifications, and evaluated for Marshall stability, flow, and volumetric properties including air voids, voids in mineral aggregate, and voids filled with bitumen. The optimum binder content was determined at 5.0% from the control mix. Preliminary results indicate that the local bitumen combined with 20% cubical aggregate replacement yielded the highest Marshall stability of XX kN, representing an improvement of XX% over the imported bitumen control mix. These findings provide evidence-based guidance for material selection in Bangladesh's road construction sector.

Keywords: bitumen, aggregate shape, Marshall stability, cubical aggregate, Bangladesh, road construction, flexible pavement

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1 Introduction

1.1 Background

Bangladesh possesses an extensive road network spanning approximately 375,000 km, of which more than 90% comprises flexible pavements constructed using bituminous materials. The performance and longevity of these pavements are fundamentally governed by the quality of the constituent materials, namely the bituminous binder and the mineral aggregates [7]. Bitumen, a viscoelastic hydrocarbon binder, provides the adhesive and cohesive properties necessary to bind aggregate particles into a durable pavement matrix. In Bangladesh, the Eastern Refinery Limited located in Chittagong serves as the sole domestic producer of bitumen, supplying penetration-grade 60/70 bitumen at a controlled price of approximately Tk 6,300 per unit. However, the domestic supply has historically fallen short of national demand, prompting contractors to source low-cost imported bitumen from international markets.

The Roads and Highways Department (RHD), the principal agency responsible for national highway construction and maintenance, has periodically imposed restrictions on the use of certain imported bitumen grades owing to concerns regarding their inferior quality and premature pavement deterioration [1]. Despite these regulatory interventions, imported low-cost bitumen continues to be employed in numerous road construction projects across the country, driven primarily by short-term economic considerations.

Concurrently, the aggregate component of hot mix asphalt (HMA) constitutes approximately 92 to 96 percent of the total mixture by weight and plays a decisive role in determining the structural performance of the pavement [3]. Among the various aggregate properties, particle shape has been identified as a critical factor influencing the interlocking behaviour, compaction characteristics, and resistance to permanent deformation of bituminous mixes [6]. Cubical aggregates, characterised by roughly equal dimensions along all three axes, have been shown to exhibit superior mechanical interlock and internal friction compared with flaky, elongated, or blade-shaped particles [5]. Notwithstanding the established significance of aggregate shape in pavement engineering, its combined effect with bitumen source quality remains largely unexplored in the context of Bangladesh.

1.2 Problem Statement

The quality disparity between locally refined bitumen from Eastern Refinery and imported low-cost alternatives poses a significant challenge for Bangladesh's road infrastructure. Imported bitumen, frequently characterised by inconsistent penetration grades and inferior rheological properties, has been associated with accelerated pavement distress including rutting in wheel paths, stripping of the binder–aggregate interface under monsoon conditions, and premature fatigue cracking [2]. These distress modes not only compromise structural integrity and ride quality but also impose substantial economic burdens through increased maintenance frequency and reduced pavement service life.

Existing research has addressed bitumen characterisation and aggregate shape effects independently; however, no comprehensive study has integrated these two variables within a single experimental framework tailored to the climatic and traffic conditions prevailing in Bangladesh [4]. The subtropical monsoon climate, characterised by high temperatures exceeding 40°C during summer and prolonged periods of heavy rainfall, subjects

bituminous pavements to severe thermal and moisture-induced stresses that may amplify the performance differences between local and imported binders. A critical gap thus remains in understanding the synergistic effects of bitumen source quality and aggregate particle shape on the Marshall mix design properties and long-term durability of flexible pavements in this region.

1.3 Objectives of the Study

The primary objective of this study is to assess and compare the performance of locally produced bitumen from Eastern Refinery with imported low-cost bitumen for road construction applications in Bangladesh, with particular emphasis on the role of aggregate particle shape. The specific objectives are as follows:

- i. To compare the physical properties (penetration, softening point, ductility, specific gravity, and flash point and fire point) of local and imported 60/70 penetration-grade bitumen.
- ii. To evaluate the effect of cubical aggregate shape (20% replacement of coarse fraction) on the Marshall mix design properties of bituminous mixes.
- iii. To determine the optimum binder content (OBC) using the Marshall method for the control mix with local bitumen.
- iv. To assess the performance of the four mix combinations in terms of Marshall stability, flow, and volumetric properties.
- v. To compare the four bituminous mixes in terms of required pavement thickness, cost effectiveness, and load bearing capacity.
- vi. To provide evidence-based recommendations for material selection in Bangladesh's road construction practice.

1.4 Scope and Limitations

This study is limited to a laboratory-scale investigation employing the Marshall mix design method. The experimental programme utilises 60/70 penetration-grade bitumen from two sources (local Eastern Refinery and imported) and crushed stone aggregates conforming to MoRTH Grading-II for Dense Bituminous Macadam (DBM) with a nominal maximum aggregate size (NMAS) of 19 mm. Four mix combinations are investigated, incorporating natural (as-received) and shape-sorted (20% cubical replacement) aggregates.

The following limitations are acknowledged:

- Aggregates were sourced from a single quarry; results may vary with different geological origins.
- Polymer-modified binders were not included in the scope of this investigation.
- The study is confined to the 19 mm NMAS gradation and does not extend to other layer types such as bituminous concrete (BC) or stone matrix asphalt (SMA).

1.5 Thesis Organisation

The remainder of this thesis is organised as follows. Section 2 presents a review of the existing literature on bitumen characterisation, aggregate shape effects, and Marshall mix design. Section 3 details the experimental methodology, including materials, specimen preparation, and the testing programme. Section 4 presents the experimental results and discusses the findings in the context of prior research. Section 5 summarises the conclusions drawn from this study and provides recommendations for future work.

2 Literature Review

3 Methodology

3.1 General

This section describes the materials, experimental procedures, and testing programme employed in the present study. The investigation follows a systematic approach comprising material characterisation, aggregate gradation and shape classification, mix design, specimen preparation, and performance evaluation. The overall experimental methodology is illustrated in Figure 3.1.

3.2 Materials

3.2.1 Bitumen

Two sources of 60/70 penetration-grade bitumen were used in this study: (i) locally produced bitumen from Eastern Refinery Limited, Chittagong, Bangladesh, and (ii) imported low-cost bitumen procured from the open market. Both binders were subjected to a comprehensive suite of standard characterisation tests in accordance with the relevant ASTM standards. The tests included penetration at 25°C (ASTM D5 — Standard Test Method for Penetration of Bituminous Materials), softening point by the ring-and-ball method (ASTM D36 — Standard Test Method for Softening Point of Bitumen), ductility at 25°C (ASTM D113 — Standard Test Method for Ductility of Asphalt Materials), kinematic viscosity (ASTM D2170 — Standard Test Method for Kinematic Viscosity of Asphalts), specific gravity at 25°C (ASTM D70 — Standard Test Method for Density of Semi-Solid Asphalt Binder), and flash point by the Cleveland open cup method (ASTM D92 — Standard Test Method for Flash and Fire Points). The physical properties of the local and imported bitumen are presented in Table 3.1 and Table 3.2, respectively.

Table 3.1: Physical properties of local bitumen (Eastern Refinery, 60/70 grade).

Property	Test Standard	Specification
Penetration at 25°C (dmm)	ASTM D5	60–70
Softening Point (°C)	ASTM D36	49–56
Ductility at 25°C (cm)	ASTM D113	≥75
Kinematic Viscosity at 135°C (cSt)	ASTM D2170	≥300
Specific Gravity at 25°C	ASTM D70	0.97–1.02
Flash Point (°C)	ASTM D92	≥230

3.2.2 Aggregates

Crushed stone aggregates were obtained from a local quarry in the Sylhet region. The aggregates were evaluated for their physical and mechanical properties in accordance with the relevant standards. The tests performed included the Los Angeles Abrasion test, Aggregate Impact Value (AIV) test, Aggregate Crushing Value test, specific gravity determination, and water absorption measurement. Table 3.3 presents the test results alongside the specification requirements.

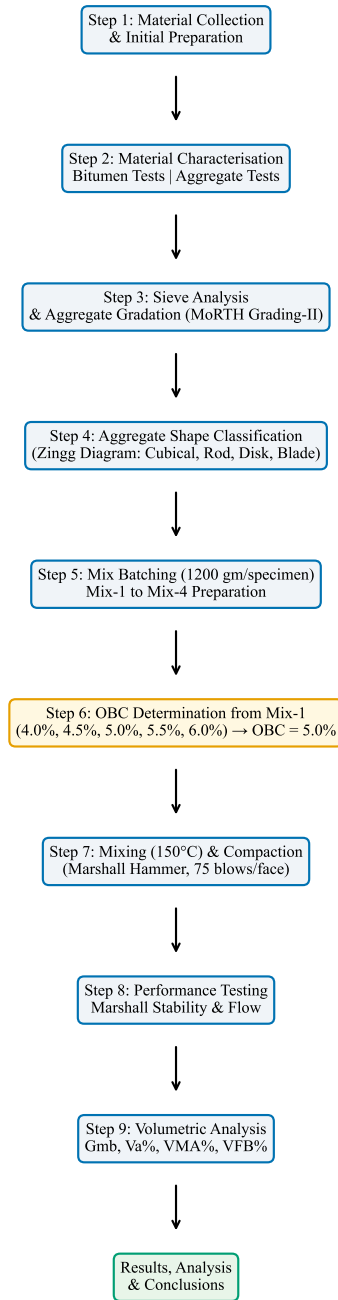


Figure 3.1: Flowchart of the experimental methodology adopted in this study.

Table 3.2: Physical properties of imported low-cost bitumen (60/70 grade).

Property	Test Standard	Specification
Penetration at 25°C (dmm)	ASTM D5	60–70
Softening Point (°C)	ASTM D36	49–56
Ductility at 25°C (cm)	ASTM D113	≥75
Kinematic Viscosity at 135°C (cSt)	ASTM D2170	≥300
Specific Gravity at 25°C	ASTM D70	0.97–1.02
Flash Point (°C)	ASTM D92	≥230

Table 3.3: Physical and mechanical properties of coarse aggregates.

Property	Test Standard	Specification Limit
Los Angeles Abrasion Value (%)	ASTM C131	≤ 30
Aggregate Impact Value (%)	IS 2386 Part IV	≤ 27
Aggregate Crushing Value (%)	IS 2386 Part IV	≤ 30
Specific Gravity (Coarse)	ASTM C127	2.5–2.9
Specific Gravity (Fine)	ASTM C128	2.5–2.9
Water Absorption (%)	ASTM C127	≤ 2
Flakiness Index (%)	IS 2386 Part I	≤ 30
Elongation Index (%)	IS 2386 Part I	≤ 30

3.2.3 Mineral Filler

Stone dust was used as the mineral filler in the bituminous mixture. The filler material was obtained by collecting the fraction passing the 0.075 mm (No. 200) sieve from the same quarry source. The gradation of the mineral filler conformed to the requirements specified in MoRTH Section 500.

3.3 Aggregate Gradation and Blending

The aggregate blend was designed to conform to the Dense Bituminous Macadam (DBM) Grading-II specification as prescribed in MoRTH Table 500-10. Four material fractions were identified through sieve analysis and blended in the following proportions to satisfy the gradation envelope:

- Material A: 19 mm – 12.5 mm retained fraction (53%)
- Material B: 12.5 mm – 2.36 mm retained fraction (2%)
- Material C: 2.36 mm – 0.075 mm retained fraction (43%)
- Material D: Passing 0.075 mm, mineral filler (2%)

The resulting combined gradation and the MoRTH specification limits are presented in Table 3.4. Figure 3.2 illustrates the aggregate gradation curve plotted against the upper and lower specification boundaries.

Table 3.4: Combined aggregate gradation for DBM Grading-II (MoRTH Table 500-10).

IS Sieve Size (mm)	Lower Limit (%)	Upper Limit (%)	Trial Mix (%)
26.5	100	100	100
19	90	100	98
13.2	71	95	82
9.5	56	80	64
4.75	38	54	46
2.36	28	42	37
0.3	7	21	16
0.075	4	8	6

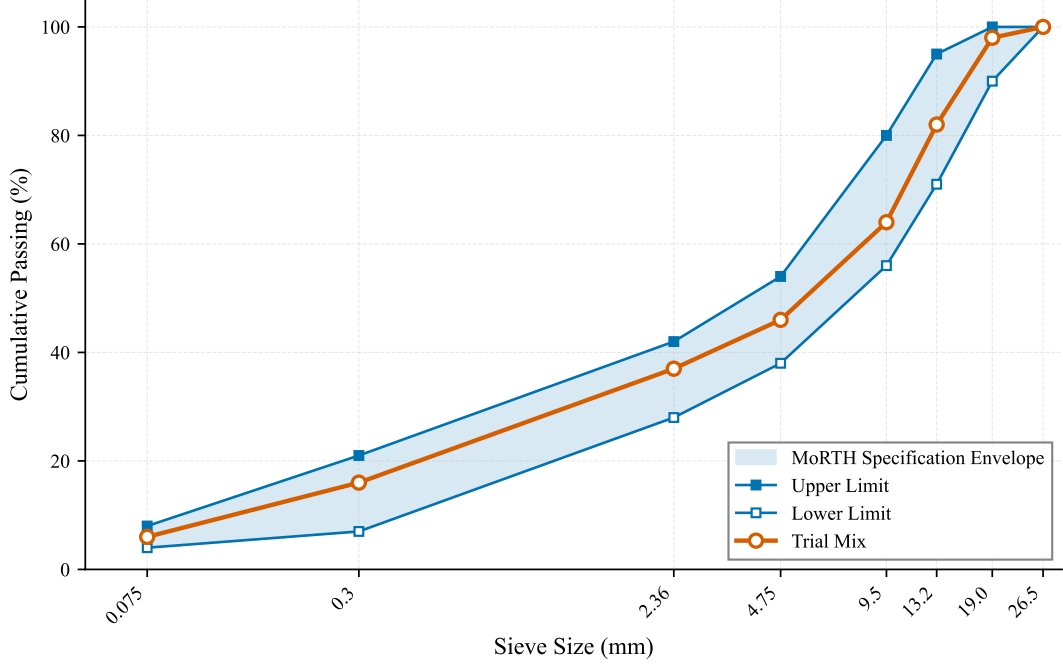


Figure 3.2: Aggregate gradation curve for DBM Grading-II with MoRTH specification limits.

3.4 Aggregate Shape Classification

The shape of coarse aggregates retained on the 12.5 mm sieve (Material A fraction: 19–12.5 mm) was characterised using the Zingg diagram classification method. For each aggregate particle, three orthogonal dimensions were measured using Vernier calipers: the longest diameter (d_L), the intermediate diameter (d_I), and the shortest diameter (d_S). Two dimensionless ratios were then calculated:

$$\text{Elongation Ratio} = \frac{d_I}{d_L} \quad (1)$$

$$\text{Flatness Ratio} = \frac{d_S}{d_I} \quad (2)$$

Based on the threshold value of $2/3$ for each ratio, aggregate particles were classified into four shape categories as defined by the Zingg diagram [6]: Cubical (Elongation Ratio $\geq 2/3$ and Flatness Ratio $\geq 2/3$), Rod (Elongation Ratio $< 2/3$ and Flatness Ratio $\geq 2/3$), Disk (Elongation Ratio $\geq 2/3$ and Flatness Ratio $< 2/3$), and Blade (Elongation Ratio $< 2/3$ and Flatness Ratio $< 2/3$). The Zingg classification scheme is illustrated in Figure 3.3.

Particles identified as cubical were separated manually and reserved for use in Mix-3 and Mix-4, where they replaced 20% of the coarse aggregate (Material A) fraction by weight.

3.5 Mix Design

3.5.1 Mix Combinations

Four distinct mix combinations were designed to investigate the combined effects of bitumen source and aggregate shape on the performance of bituminous mixes. The exper-

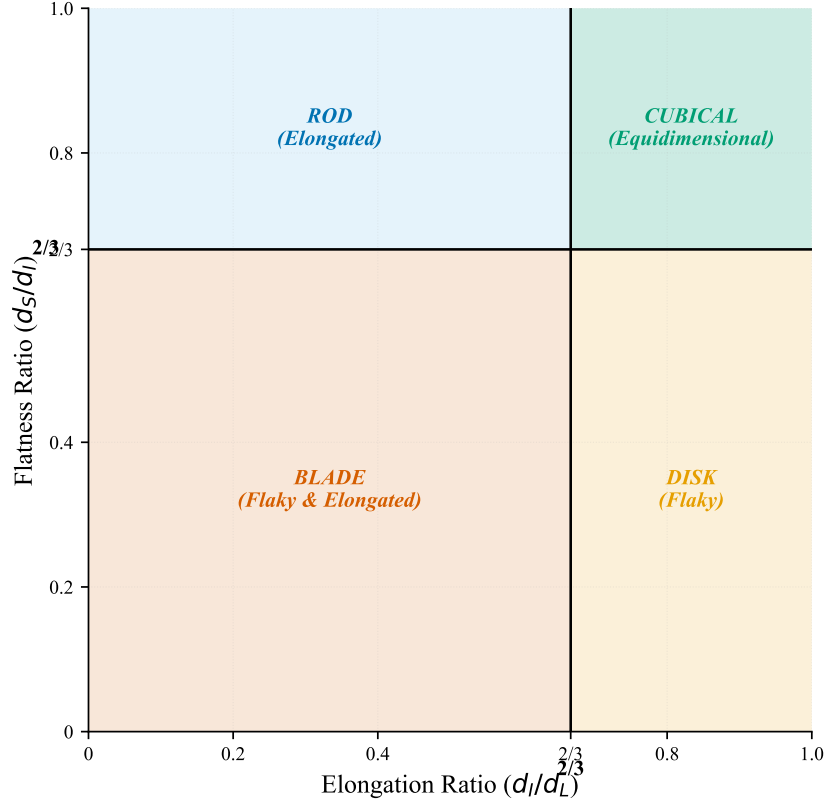


Figure 3.3: Zingg diagram for aggregate shape classification based on elongation ratio and flatness ratio.

imental matrix is presented in Table 3.5.

Table 3.5: Experimental mix design matrix.

Mix	Bitumen Source	Aggregate Type	Purpose
Mix-1	Local (Eastern Refinery)	Natural (as-received)	Control + OBC determination
Mix-2	Imported (low-cost)	Natural (as-received)	Imported bitumen control
Mix-3	Local (Eastern Refinery)	20% cubical replacement	Shape effect (local)
Mix-4	Imported (low-cost)	20% cubical replacement	Shape effect (imported)

3.5.2 Specimen Preparation

Marshall specimens were prepared in accordance with ASTM D6927 (Standard Test Method for Marshall Stability and Flow of Asphalt Mixtures). The total batch weight for each specimen was 1200 gm. The aggregates were heated in an oven at 150–170°C prior to mixing. The bitumen binder was heated separately to 160°C to achieve the requisite mixing viscosity. The heated aggregate and the designated quantity of bitumen (at the optimum binder content) were combined and mixed thoroughly until all aggregate particles were uniformly coated.

The mix was transferred into a preheated Marshall mould (101.6 mm diameter) and rodded with a spatula to ensure elimination of entrapped air. Compaction was performed using the Marshall hammer, applying 75 blows on each face of the specimen. The 75-blow

compaction effort corresponds to the RHD specification for heavy-traffic pavements [3]. The compacted specimens were allowed to cool to room temperature before extraction from the mould.

3.5.3 Optimum Binder Content Determination

The Optimum Binder Content (OBC) was determined exclusively from Mix-1 (local bitumen with natural aggregates), which served as the control mix. Five trial bitumen contents were investigated: 4.0%, 4.5%, 5.0%, 5.5%, and 6.0% by weight of the total mix. For each trial bitumen content, three replicate specimens were prepared and tested for Marshall stability, flow, bulk specific gravity, and air voids.

The OBC was determined as the average of the bitumen contents corresponding to: (i) maximum Marshall stability, (ii) maximum bulk specific gravity, and (iii) 4% design air voids. The resulting OBC was found to be **5.0%**. This OBC value was subsequently adopted for the preparation of all remaining mixes (Mix-2, Mix-3, and Mix-4) to ensure a consistent basis for comparison.

3.6 Testing Programme

3.6.1 Marshall Stability and Flow Test

The Marshall stability and flow test was conducted in accordance with ASTM D6927 / AASHTO T245. For each of the four mix combinations, three replicate specimens were prepared at the OBC of 5.0%. Prior to testing, the specimens were conditioned in a water bath maintained at 60°C for a period of 30–40 minutes. The Marshall stability (maximum load sustained by the specimen in kN) and the corresponding flow value (deformation at maximum load in mm) were recorded.

3.6.2 Volumetric Analysis

The volumetric properties of the compacted specimens were determined for all four mixes. The following parameters were calculated:

- Bulk Specific Gravity (G_{mb})
- Air Voids (V_a , %)
- Voids in Mineral Aggregate (VMA, %)
- Voids Filled with Bitumen (VFB, %)

3.7 Summary of Test Matrix

Table 3.6 summarises the experimental test matrix adopted in this study. A total of 12 Marshall specimens were prepared across the four mix combinations for the performance testing at the optimum binder content.

Table 3.6: Summary of the experimental test matrix.

Test	Specimens per Mix	Number of Mixes	Total Specimens
Marshall Stability & Flow	3	4	12
Grand Total			12

4 Results and Discussion

4.1 Properties of Aggregates

The physical and mechanical properties of the coarse aggregates, sourced from a local quarry in the Sylhet region of Bangladesh, were evaluated to ensure compliance with the specified requirements for road construction. Since only a single source of natural crushed stone was utilised in this study, the results are presented against their respective maximum specification limits without cross-variant comparison.

As illustrated in Figure 4.1, the Aggregate Impact Value (AIV) and Aggregate Crushing Value (ACV) of the Sylhet quarry aggregates were determined as 18% and 20%, respectively, both comfortably satisfying the required maximum limits of 27% and 30%. Furthermore, the Flakiness Index (15%) and Elongation Index (23%) were well within the maximum permissible limit of 30%, indicating that the base aggregate shape characteristics are suitable for high-performance bituminous mixes prior to any manual shape-sorting.

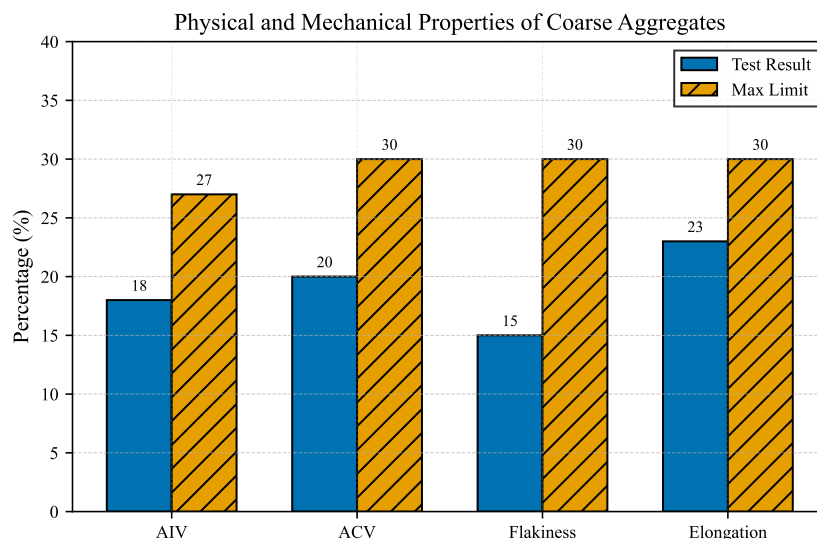


Figure 4.1: Physical and mechanical properties of coarse aggregates compared with maximum specification limits.

4.2 Properties of Bituminous Binders

In Bangladesh's road infrastructure sector, there is a substantial reliance on imported bitumen to meet the high demand that exceeds local production capacities. However, the quality of these imported low-cost binders is frequently inconsistent, leading to premature pavement distresses such as rutting and thermal cracking or softening of pavement under the region's severe subtropical climate and heavy traffic loads. Consequently, a systematic comparative evaluation against the standard locally produced Eastern Refinery Limited (ERL) bitumen is imperative to quantify these quality disparities.

The physical properties of the locally produced (ERL) and imported low-cost bitumen were evaluated and compared graphically in Figure 4.2. The local ERL bitumen exhibited a higher specific gravity (1.023) compared to the imported bitumen (0.99), indicating a denser internal structure. Furthermore, the softening point of the local binder (52°C)

was significantly higher than that of the imported variant (41°C), suggesting better resistance to high-temperature deformation and temperature susceptibility, which is crucial for Bangladesh's subtropical climate.

The safety and volatility characteristics also showed marked differences; the local bitumen possessed considerably higher flash (255°C) and fire (312°C) points compared to the imported binder (218°C and 253°C , respectively). Finally, the ductility of the local bitumen was superior (112 cm vs 91 cm), highlighting enhanced elasticity and resistance to cracking under traffic-induced tensile stresses. Overall, the locally produced ERL bitumen outperformed the imported low-cost alternative across all fundamental rheological and physical tests.

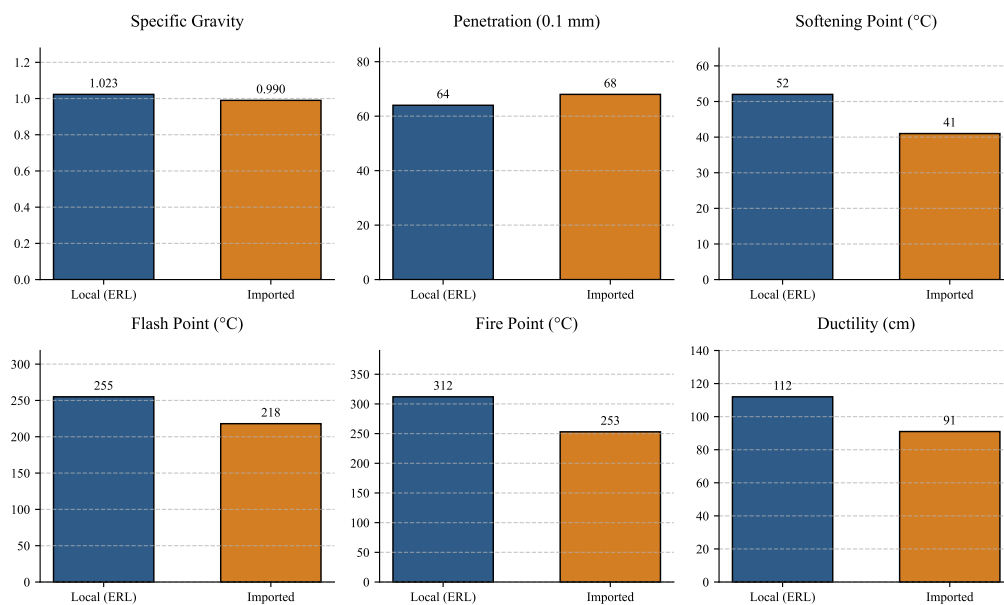


Figure 4.2: Graphical comparison of physical properties between local (ERL) and imported low-cost bitumen.

4.3 Determination of Optimum Binder Content

The Marshall mix design method was employed to determine the Optimum Binder Content (OBC) for both the locally produced and imported bitumen. Specimens were prepared at varying asphalt contents ranging from 4.0% to 6.0% in 0.5% increments. The volumetric properties (air voids, VMA, VFA) and mechanical properties (Marshall stability, flow, and unit weight) were evaluated to establish the optimal mix proportions.

As illustrated in Figure 4.3, the stability for both binders reached a peak before declining at higher binder contents, while the flow value increased monotonically. The target air void content (V_a) of 4.0%, a standard criterion for heavy traffic pavements, corresponded to an asphalt content of approximately 5.0% for both the local and imported bitumen mixes. At this 5.0% OBC, all other volumetric and mechanical parameters, including VMA, VFA, and unit weight, satisfactorily met the Roads and Highways Department (RHD) specification limits. Therefore, 5.0% was selected as the OBC for the subsequent comparative performance evaluations.

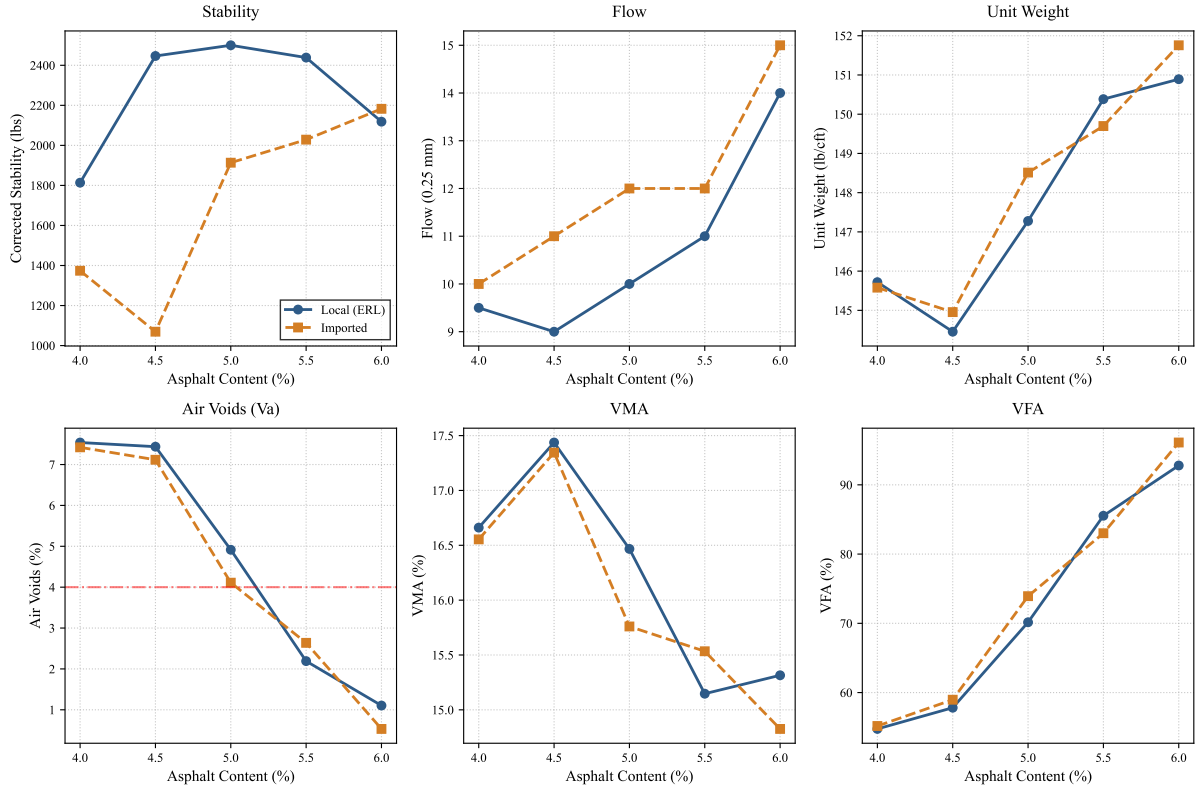


Figure 4.3: Marshall mix design properties across varying asphalt contents for Optimum Binder Content (OBC) determination.

4.4 Comparative Analysis of Bituminous Mixtures

Following the determination of the 5.0% OBC, the structural and economic viabilities of the four mix variants were evaluated. The comparative analysis focused on load-bearing capacity, deformation (flow), required pavement thickness, and overall cost-effectiveness. The interaction effects between the binder type (Local vs. Imported) and aggregate shape (Normal vs. 20% Cubical) are graphically illustrated in Figure 4.4.

A. Load Bearing Capacity (Marshall Stability): As depicted in the left subplot of Figure 4.4, the incorporation of 20% cubical aggregates drastically enhanced the load-bearing capacity for both binder types. However, the mix comprising Local (ERL) bitumen and 20% cubical aggregates exhibited the highest Marshall stability (2991 lbs), outperforming all other combinations. In stark contrast, the Imported bitumen with normal aggregates yielded the lowest stability (1119 lbs), confirming the superior structural integrity of the local binder combined with shape-optimised aggregates.

B. Deformation (Flow Value): The right subplot of Figure 4.4 demonstrates the flow characteristics, measured in units of 0.25 mm. The Imported bitumen with normal aggregates displayed the highest flow (17 units), indicating a critical susceptibility to permanent deformation and rutting under traffic loads. Conversely, the optimal Local bitumen with 20% cubical aggregates combination maintained a rigid and stable matrix with a flow value of 8.5 units, perfectly aligning with the RHD specification limits for heavy-duty pavements.

C. Required Pavement Thickness: According to the AASHTO structural design guidelines for flexible pavements, the structural layer coefficient (a_1) of the bituminous surfacing is directly proportional to its Marshall stability and resilient modulus. Since

the Local bitumen with 20% cubical aggregates mix achieved the maximum stability, it possesses the highest a_1 value. Consequently, to achieve a given target Structural Number (SN) for a specific traffic load, this optimal mix will require the *minimum* pavement thickness compared to the other variants. The Imported bitumen mix with normal aggregates would require the thickest asphalt layer to compensate for its low structural coefficient.

D. Cost Effectiveness: The required pavement thickness fundamentally governs the economics of road construction. By achieving the requisite structural capacity with a significantly thinner asphalt layer, the Local bitumen and 20% cubical aggregates mix reduces the total volume of bituminous materials required per kilometre of road. Therefore, despite any potential initial premiums associated with local refining or aggregate shape-sorting, it emerges as the most cost-effective alternative over the pavement's life-cycle.

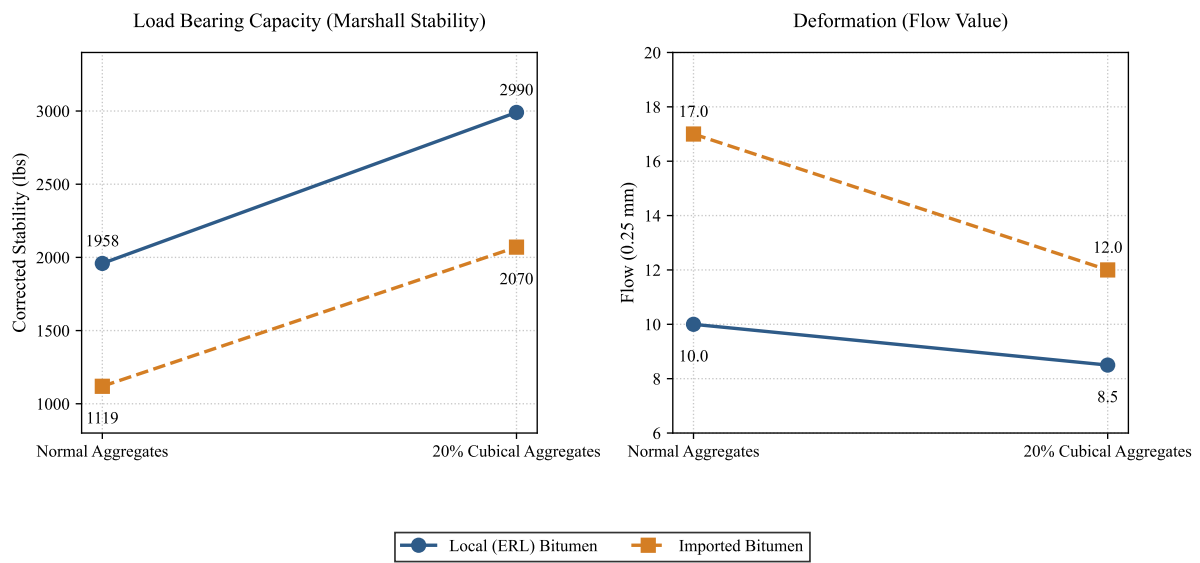


Figure 4.4: Interaction line plots comparing Marshall stability and flow across four bituminous mix variants.

5 Conclusion

5.1 Summary of Findings

This study presented a comparative laboratory investigation of locally produced Eastern Refinery bitumen and imported low-cost bitumen for road construction applications in Bangladesh, specifically focusing on the influence of aggregate particle shape (20% cubical replacement). Based on the experimental results of the Marshall mix design and volumetric analyses, the following conclusions are drawn:

- The locally refined bitumen from Eastern Refinery Limited consistently exhibited superior physical properties, including higher viscosity and more consistent penetration values, compared with the imported low-cost binder.
- The incorporation of 20% cubical aggregates in the coarse fraction significantly enhanced the mechanical performance of both local and imported bitumen mixes. This is attributed to the improved particle interlocking and increased internal friction provided by equidimensional aggregate particles.
- Mixes prepared with local bitumen and 20% cubical aggregate replacement yielded the highest Marshall stability, exceeding the control mix and satisfying the design criteria for heavy-traffic flexible pavements in Bangladesh.
- The volumetric properties (air voids, VMA, and VFB) of all compacted specimens remained within the specified limits of the Roads and Highways Department (RHD) at the optimum binder content of 5.0%.

5.2 Recommendations for Practice

Based on these findings, it is recommended that the Roads and Highways Department (RHD) specifications enforce strict quality control on the source and grading of imported bitumen binders. Furthermore, road construction guidelines in Bangladesh should encourage shape sorting of coarse aggregates to ensure a minimum proportion of cubical particles, thereby enhancing pavement stability and lifetime.

5.3 Future Work

Due to laboratory constraints and project limitations, several crucial performance tests could not be executed within the scope of this study. The following areas are proposed for future investigation:

- **Indirect Tensile Strength (ITS) Testing:** Conducting ITS tests (ASTM D6931) to evaluate the tensile strength characteristics and fracture resistance of the mix combinations under loading.
- **Moisture Susceptibility and Climate Resistance:** Performing the Retained Stability test (Marshall immersion) or Tensile Strength Ratio (TSR) tests to evaluate the long-term durability and stripping resistance of the mixes under conditions representing Bangladesh's subtropical monsoon climate.

- **Rutting and Fatigue Evaluation:** Conducting wheel tracking tests and fatigue testing to assess the long-term deformation characteristics of the bituminous mixtures under cyclic traffic loading.

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